

EECS 293
Software Craftsmanship
2016 Fall Semester

Quiz 1

Your name: _____

Your discussion section (time and location, or section number):

Check here if your discussion session is on Wednesday, 2:15pm ☐

The test is 15 minutes, 10 points
Closed books and notes

1. (3 points) State whether error handling should be treated as an architectural issue or as a coding convention, and briefly explain your reasoning.

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Quiz 2

Your name: _____

Your discussion leader: _____

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The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Briefly explain the benefits of writing a comment to explain a missing else clause.

2. (3 points) Identify and fix the problem with the following code snippet:

```
if (tokenTypeFound == true) {  
    ...  
} else {  
    ...  
}
```

3. (4 points) The Student Disinformation System uses variables that represent university course offerings. These variables are strings consisting of course subject, catalog number, group (for example, engineering or management), and campus building, separated by commas. The format is pervasive throughout the code, and is used for local variables, and for arguments and return values of many methods. Briefly explain the problems with this approach and suggest an alternative.

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Quiz 3

Your name: _____

Your discussion leader: _____

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The test is 10 minutes, 10 points
Closed books and notes

1. (7 points) Calculate the cyclomatic complexity of the following code snippets. Start the count at 0. Show your work.

a.

```
...
switch (errorCode) {
case TRAILING_INPUT:
    msg = "unexpected trailing input";
    break;

case TOKEN_EXPECTED:
    while (!(leftover > 10)) {
        leftover += a[i];
    }
    break;

default:
    msg = "unspecified";
}
...
```

b.

```
...
MAIN_LOOP:
do {
    if (sum > MIN_SUM)
        break MAIN_LOOP;
    else if (sum < MAX_SUM)
        sum += a[i] > b[i] ? a[i] + b[i] : a[i] * b[i];
    else
        i++;
    i += sum;
} while (i < rows)
...
```

2. (3 points) Rewrite the following code snippet to remove the duplicate invocation to `matcher.find()`.

```
...
matcher.find();
while (matcher.group(tokenType).equals("and")) {
    parse(inputString);
    matcher.find();
}
...
```

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Quiz 4

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The test is 10 minutes, 10 points
Closed books and notes

1. (4 points) Calculate the cyclomatic complexity of the following code snippet. Start the count at 0. Show your work.

```
...
if (sum > MAX_SUM || i >= n) then
    return;
else {
    MAIN_LOOP:
    while (sum < MAX_SUM) {
        sum = a[i];
        i++;
        break MAIN_LOOP;
    }
}
...
```

2. (2 points) Give a short example of a contract for a class interface.

3. (4 points) State whether the programmer has coupled bar and BaseClass tightly or loosely, give a brief explanation of your answer, and suggest a possible fix.

```
class BaseClass {
    void double foo() {
        return 0.;
    }
}

class DerivedClass extends BaseClass {
    ...
    void double foo() {
        return 1.;
    }
    ...
}

class Klass {
    void bar(BaseClass obj) {
        // Programmer's note:
        // I know that obj always belongs
        // to DerivedClass
        // Therefore, no exception thrown.
        return 1. / obj.foo();
    }
}
```

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Quiz 5

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Closed books and notes

1. (8 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
 - a. Adds the argument (an airplane) to the list of vehicles parked at this hangar.
 - b. Returns the maximum fuel (lb.), a private variable of this airplane.
 - c. Calculates and returns the maximum payload (lb.) for this airplane.

- d. Calculates and returns the maximum range (in nautical miles) of this airplane.
 - e. Adds a payload (lb.) to this airplane and returns true if the operation succeeded and false otherwise (for example, if the maximum payload would be exceeded).
 - f. Returns whether this airplane has been cleared for takeoff.
 - g. Convert knots to feet per second.
 - h. Displays an icon representing this airplane.
2. (2 points) Briefly explain the problem with the argument of following routine, and propose a fix.

```
int foo(int x) {  
    x = Math.sin(x);  
    x = Math.cos(x) + this.location;  
    ... 100 lines ...  
    return x;  
}
```

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Quiz 6

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The test is 10 minutes, 10 points
Closed books and notes

1. (4 points) The following two lines of pseudo-code are supposed to convey the same concept. Circle the most appropriate pseudo-code and briefly explain your answer.

$y \leftarrow \tan(x)$

`y = Math.tan(x);`

2. (3 points) State whether the following routine satisfies its pre-conditions and post-conditions and briefly explain your answer.

```
/*      Pre-condition:  $x > 0$ 
      Post-condition: return value is  $1/\text{sqrt}(x)$  */
double invSqrt(double x) {
    return Math.sqrt(x) / x;
}
```

3. (3 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
- Mark as completed the assignment passed as an argument, and returns a status code.
 - Mark this assignment as completed and returns a status code.
 - Returns whether this assignment has been completed by the date passed as an argument. The return value must be calculated since this assignment has no variable to hold the requested information.

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Quiz 7

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Closed books and notes

1. (7 points) Write test cases for the following code snippet. The test cases should cover code coverage, branch coverage, and both simple and compound boundaries. Other categories are not required. Show your work, and in particular label each test case as to whether it addresses code coverage, branch coverage, simple boundaries, compound boundaries, or a combination. You should outline the conditions of each test and can omit the specific JUnit code. You are not allowed to modify the code under test. You can assume that `x`, `y`, and `maxValue` are integer variables.

```
...  
    if (x < y && y < maxValue) {  
        ...
```

2. (3 points) A manager has collected the following statistics about a software project. The project is now in its third year, and consists of twenty classes. The team has found and corrected 41 defects, 20 of which in a single class called Stage, and the rest spread out more or less evenly among the remaining nineteen classes.

Choose one of the following courses of action, and briefly explain your choice. Many but not all of the following choices are reasonable, and it is sufficient to discuss only one of them.

- a. Reevaluate the architecture of Stage and if needed refactor or replace Stage.
- b. Reevaluate the architecture of the remaining 19 classes so that they are more tightly coupled with Stage.
- c. Reevaluate the pseudo-code for Stage's methods and refactor those methods as needed.
- d. Generate a coverage report of Stage and if needed add tests to ensure 100% coverage.
- e. Shift the focus to the testing and debugging of the remaining 19 classes.
- f. Scrap the project.

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Quiz 8

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Closed books and notes

1. The following questions are about the notion of *failure atomicity*.
 - a. (2 points) Briefly explain the concept of failure atomicity.

 - b. (2 points) Suggest a method to enforce failure atomicity and briefly justify your answer.

2. (2 points) Briefly define the *singleton pattern*. (No implementation is required.)

3. (4 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
- a. Remove one segment from this digit, and return a Boolean denoting whether the operation succeeded.
 - b. Return a 3×3 String array representing a specific digit from the String passed as an argument, or an empty optional if no such digit exists.
 - c. Returns the location of this digit in the input string. The location is a private variable of this digit.
 - d. Convert this digit to a Roman numeral.

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Quiz 9

Your name: _____

Your discussion leader: _____

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The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Give an example of a public mutable companion class to an immutable class, and briefly outline a scenario where it is useful.
2. (2 points) Briefly explain the issues that arise if a public non-final class has a protected method.

3. (3 points) Among the following potential class names, circle the one that is inappropriate and briefly explain your answer:
- a. OrganizingModule
 - b. Organization
 - c. Organize
 - d. OrganizedWireframe
4. (2 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
- a. Return the number of annotations of this wireframe element (calculated as the size of a private list of annotations).
 - b. Add an annotation to this wireframe element, and returns the number of annotations on this wireframe element.

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Quiz 10

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Closed books and notes

1. (4 points) Give an example in Java of a class self-use of over-ridable methods, including the relevant documentation. (Minor syntax errors are unimportant.).

2. (4 points) Briefly explain the difference between containment and aggregation and give a short example.

3. (2 points) Suppose that a class has 21 private methods. Briefly argue that the class is poorly designed and propose an alternative.

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Quiz 11

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Closed books and notes

1. (3 points) A company memo proposes that programmers should manually type blank spaces in their comments so that the comments are snugly justified to a width of 80 characters. Briefly explain at least two problems that the new policy will create.
2. (2 points) Briefly outline a problem with comments placed at the end of the line.

3. (2 points) State whether the following routine is a function or a procedure and give an appropriate name to the routine. The keyword “this” refers to the object to which the routine belongs.

Locks the wireframe widget contained in this wireframe canvas at the (x, y) coordinates passed as an argument, and returns a boolean denoting whether the operation was successful.

4. (3 points) Calculate the cyclomatic complexity of the following code snippet. Start the count at 0. Show your work.

```
...
switch (option) {
case PRINT:
    PRINT_LOOP:
    for (Record record : records) {
        if (print(record) == COMPLETED) {
            break PRINT_LOOP;
        }
    }

default:
    logger.info("Unsupported option");
}
...
```